

F-Test for Equality of Two Population Variances (Variance Ratio Test)

F-Test (Variance Ratio Test)

F-Test is a statistical hypothesis test used to compare the **variances** of two independent populations.

It helps answer questions such as:

- Do two machines produce products with the same consistency?
- Are two horses equally consistent in running performance?
- Is the variability in marks of two classes the same?

The test compares the ratio of two sample variances.

Conditions for Applying F-Test

The F-test can be applied when:

- Two independent random samples are selected.
- Both populations are normally distributed.
- Samples are independent.
- Data are quantitative (continuous).

Notation

Let

- (X_1) = Observations from Population 1
- (X_2) = Observations from Population 2

Sample sizes

- (n_1)
- (n_2)

Sample variances

- (S_1^2)
- (S_2^2)

Symbol	Meaning	Explanation
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(H_0)	Null Hypothesis	The initial assumption that there is no difference between the two population variances.
(σ)	Population Standard Deviation	Represents the spread (dispersion) of values in an entire population. The Greek letter sigma (σ) is used because it refers to a population parameter.
(σ^2)	Population Variance	Variance is the square of the standard deviation. It measures how much the data is spread around the population mean.
(σ_1^2)	Variance of Population 1	The true variance of the first population. It is usually unknown and estimated using the sample variance (s_1^2).
(σ_2^2)	Variance of Population 2	The true variance of the second population. It is also unknown and estimated using the sample variance (s_2^2).
$(=)$	Equal To	Indicates that the null hypothesis assumes the two population variances are equal.

Step 1: State the Hypotheses

Null Hypothesis

$$[H_0: \sigma_1^2 = \sigma_2^2]$$

The two populations have the **same variance**.

Alternative Hypothesis

Two-tailed Test

$$[H_1: \sigma_1^2 \neq \sigma_2^2]$$

The variances are different.

Right-tailed Test

$$[H_1: \sigma_1^2 > \sigma_2^2]$$

Population 1 has greater variance.

Left-tailed Test

$$[H_1: \sigma_1^2 < \sigma_2^2]$$

Population 2 has greater variance.

Step 2: Calculate Sample Variances

The sample variance is

$$\left[S^2 = \frac{1}{n-1} \left[\sum X^2 - \frac{(\sum X)^2}{n} \right] \right]$$

where

- (n)=sample size
- $(\sum X)$ =sum of observations
- $(\sum X^2)$ =sum of squared observations

Formula for Population 1

$$\left[S_1^2 = \frac{1}{n_1-1} \left[\sum X_1^2 - \frac{(\sum X_1)^2}{n_1} \right] \right]$$

Formula for Population 2

$$\left[S_2^2 = \frac{1}{n_2-1} \left[\sum X_2^2 - \frac{(\sum X_2)^2}{n_2} \right] \right]$$

Step 3: Calculate the F Statistic

Always place the **larger variance in the numerator**.

$$\left[F = \frac{\text{Larger Sample Variance}}{\text{Smaller Sample Variance}} \right]$$

This ensures

$$[F \geq 1]$$

Step 4: Determine the Level of Significance

Common choices are

- 5% (0.05)
- 1% (0.01)

Step 5: Calculate Degrees of Freedom

If

- Numerator variance belongs to Sample 1

$$[df_1 = n_1 - 1]$$

- Denominator variance belongs to Sample 2

$$[df_2 = n_2 - 1]$$

Otherwise, reverse the order according to whichever sample variance is placed in the numerator.

Step 6: Find Critical Value

Using the F-table,

Find

$$[F_{\alpha}(df_1, df_2)]$$

Example

$$[F_{0.05}(5,6) = 4.39]$$

Step 7: Decision Rule

If

$$[F_{cal} > F_{tab}]$$

Reject (H_0)

The variances are significantly different.

If

$$[F_{cal} \leq F_{tab}]$$

Accept (or fail to reject)(H_0)

The variances are statistically equal.

		F-table of Critical Values of $\alpha = 0.05$ for F(df1, df2)																		
		DF1=1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
DF2=1		161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2		18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3		10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4		7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5		6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6		5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7		5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8		5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9		5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10		4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11		4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12		4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13		4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14		4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15		4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16		4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17		4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18		4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19		4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20		4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21		4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22		4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23		4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24		4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25		4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26		4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27		4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28		4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29		4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30		4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40		4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60		4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120		3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞		3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Standard Procedure

Step 1

Write hypotheses.

Step 2

Prepare calculation table.

$$(X_1) (X_1^2) (X_2) (X_2^2)$$

Calculate

- $(\sum X_1)$
- $(\sum X_1^2)$
- $(\sum X_2)$
- $(\sum X_2^2)$

Step 3

Calculate

$$S_1^2$$

Step 4

Calculate

$$S_2^2$$

Step 5

Compute

$$F = \frac{\text{Larger Variance}}{\text{Smaller Variance}}$$

Step 6

Find

- Level of significance
- Degrees of freedom
- Critical F value

Step 7

Compare

$$F_{cal}$$

and

$$F_{tab}$$

Step 8

Write the conclusion in words.

Example: F-Test for Equality of Two Population Variances

Given Data

Horse A

28, 30, 32, 33, 33, 29, 34

Horse B

29, 30, 30, 24, 27, 29

Step 1: State the Hypotheses

The F-test is used to determine whether the two population variances are equal.

Null Hypothesis

$$H_0: \sigma_1^2 = \sigma_2^2$$

There is **no significant difference** between the variances of Horse A and Horse B.

Alternative Hypothesis

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

There is **a significant difference** between the variances.

Step 2: Level of Significance

Assume

$$\alpha = 0.05$$

This means there is a **5% risk** of rejecting the null hypothesis when it is true.

Step 3: Calculate the Sample Means

Horse A

Number of observations

$$n_1 = 7$$

Sum of observations

$$28 + 30 + 32 + 33 + 33 + 29 + 34 = 219$$

Mean

$$\bar{X}_1 = \frac{219}{7} = 31.29$$

Horse B

Number of observations

$$n_2 = 6$$

Sum

$$29 + 30 + 30 + 24 + 27 + 29 = 169$$

Mean

$$\bar{X}_2 = \frac{169}{6} = 28.17$$

Step 4: Calculate Sample Variance of Horse A

Formula

$$S^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

Observation (X)	Mean (31.29)	(X - $\bar{X}$)	((X - $\bar{X}$) ²)
28	31.29	-3.29	10.82
30	31.29	-1.29	1.65
32	31.29	0.71	0.51
33	31.29	1.71	2.94
33	31.29	1.71	2.94
29	31.29	-2.29	5.22
34	31.29	2.71	7.37

Sum of squared deviations

$$= 31.45$$

Variance

$$S_1^2 = \frac{31.45}{7 - 1}$$

$$= \frac{31.45}{6}$$

$$= 5.24 \approx 5.23$$

Therefore,

$$S_1^2 = 5.23$$

Step 5: Calculate Sample Variance of Horse B

Mean: 28.17

Observation (X)	Mean	$(X - \bar{X})$	$((X - \bar{X})^2)$
29	28.17	0.83	0.69
30	28.17	1.83	3.36
30	28.17	1.83	3.36
24	28.17	-4.17	17.39
27	28.17	-1.17	1.36
29	28.17	0.83	0.69

Sum of squared deviations
=26.85

Variance

$$S_2^2 = \frac{26.85}{6 - 1}$$

$$= \frac{26.85}{5}$$

$$= 5.37$$

Therefore,

$$S_2^2 = 5.37$$

Step 6: Calculate the F-Test Statistic

The larger variance is placed in the numerator.

Since

$$5.37 > 5.23$$

Use

$$F = \frac{5.37}{5.23}$$

$$F = 1.027$$

Therefore,

$$\boxed{F_{cal} = 1.03}$$

Step 7: Determine Degrees of Freedom

For the numerator (larger variance)

$$df_1 = n_2 - 1$$

$$= 6 - 1 = 5$$

For the denominator

$$df_2 = n_1 - 1$$

$$= 7 - 1 = 6$$

Thus,

$$df_1 = 5, \quad df_2 = 6$$

Step 8: Find the Critical Value

From the **F-distribution table** at the **5% significance level**:

- Numerator degrees of freedom = 5
- Denominator degrees of freedom = 6

The critical value is approximately:

$$\boxed{F_{0.05}(5,6) \approx 4.39}$$

If performing a strict two-tailed F-test, the rejection regions are in both tails; many introductory courses compare the larger sample variance to the upper-tail critical value after placing the larger variance in the numerator.

Step 9: Decision Rule

Reject (H_0) if

$$F_{cal} > F_{critical}$$

Otherwise,

Do not reject (H_0).

Step 10: Compare

Calculated value

$$F_{cal} = 1.03$$

Critical value

$$F_{critical} = 4.39$$

Since

$$1.03 < 4.39$$

we **do not reject** the null hypothesis.

Step 11: Conclusion

There is **no significant difference** between the variances of Horse A and Horse B at the **5% significance level**.

Hence, we conclude that the population variances can be considered **equal**.

Practice Examples (F-Test for Equality of Variances)

Example 1: Machine Consistency

Two machines produce metal rods. The diameters (mm) of randomly selected rods are:

Machine A:

50, 49, 52, 51, 50, 48, 49, 51

Machine B:

51, 54, 49, 53, 50, 52, 55, 48

Task: Test at the 5% significance level whether the two machines have the same variability in rod diameter.

Example: F-Test for Equality of Two Population Variances

Given Data

Machine A

50, 49, 52, 51, 50, 48, 49, 51

Machine B

51, 54, 49, 53, 50, 52, 55, 48

Step 1: State the Hypotheses

The F-test is used to determine whether the variances of the two populations are equal.

Null Hypothesis

$$H_0: \sigma_1^2 = \sigma_2^2$$

There is **no significant difference** between the variances of Machine A and Machine B.

Alternative Hypothesis

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

There is a **significant difference** between the variances of the two machines.

Step 2: Level of Significance

Assume

$$\alpha = 0.05$$

Step 3: Calculate the Sample Means

Machine A

Number of observations

$$n_1 = 8$$

Sum of observations

$$50 + 49 + 52 + 51 + 50 + 48 + 49 + 51 = 400$$

Mean

$$\bar{X}_1 = \frac{400}{8} = 50$$

Machine B

Number of observations

$$n_2 = 8$$

Sum of observations

$$51 + 54 + 49 + 53 + 50 + 52 + 55 + 48 = 412$$

Mean

$$\bar{X}_2 = \frac{412}{8} = 51.5$$

Step 4: Calculate Sample Variance of Machine A

Formula

$$S^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

Observation (X)	Mean (50)	$(X - \bar{X})$	$((X - \bar{X})^2)$
50	50	0	0
49	50	-1	1
52	50	2	4
51	50	1	1
50	50	0	0
48	50	-2	4
49	50	-1	1
51	50	1	1

Sum of Squared Deviations

$$0 + 1 + 4 + 1 + 0 + 4 + 1 + 1 = 12$$

Sample Variance

$$\begin{aligned}
 S_1^2 &= \frac{12}{8 - 1} \\
 &= \frac{12}{7} \\
 &= 1.714
 \end{aligned}$$

Therefore,

$$S_1^2 = 1.71$$

Step 5: Calculate Sample Variance of Machine B

Mean

$$\bar{X}_2 = 51.5$$

Observation (X)	Mean (51.5)	$(X - \bar{X})$	$((X - \bar{X})^2)$
51	51.5	-0.5	0.25
54	51.5	2.5	6.25
49	51.5	-2.5	6.25
53	51.5	1.5	2.25
50	51.5	-1.5	2.25
52	51.5	0.5	0.25
55	51.5	3.5	12.25
48	51.5	-3.5	12.25

Sum of Squared Deviations

$$0.25 + 6.25 + 6.25 + 2.25 + 2.25 + 0.25 + 12.25 + 12.25 = 42$$

Sample Variance

$$\begin{aligned} S_2^2 &= \frac{42}{8 - 1} \\ &= \frac{42}{7} \\ &= 6 \end{aligned}$$

Therefore,

$$S_2^2 = 6.00$$

Step 6: Calculate the F-Test Statistic

The larger variance is placed in the numerator.

Since

$$6.00 > 1.71$$

$$F = \frac{6.00}{1.71} = \frac{6}{12/7} = \frac{42}{12} = 3.50$$

Therefore,

$$F_{cal} = 3.50$$

Step 7: Determine the Degrees of Freedom

Since the larger variance belongs to **Machine B**,

Numerator

$$df_1 = n_2 - 1 = 8 - 1 = 7$$

Denominator

$$df_2 = n_1 - 1 = 8 - 1 = 7$$

Therefore,

$$df_1 = 7, \quad df_2 = 7$$

Step 8: Find the Critical Value

From the **F-distribution table** at the **5% significance level**:

- Numerator degrees of freedom = 7
- Denominator degrees of freedom = 7

The upper-tail critical value is approximately

$$F_{critical} \approx 3.79$$

Step 9: Decision Rule

- Reject (H_0) if

$$F_{cal} > F_{critical}$$

- Otherwise, do not reject (H_0).

Step 10: Compare

Calculated F-value

$$F_{cal} = 3.50$$

Critical F-value

$$F_{critical} = 3.79$$

Since

$$3.50 < 3.79$$

we **do not reject** the null hypothesis.

Step 11: Conclusion

At the **5% significance level**, there is **no significant difference** between the variances of Machine A and Machine B.

Therefore, the production variability of the two machines can be considered **equal**.

Example 2: Examination Scores

Two sections of students obtained the following marks:

Section A:

68, 72, 74, 70, 69, 71, 73

Section B:

62, 80, 75, 69, 77, 65, 82

Task: Use an F-test to determine whether the two sections have equal variance in marks.

Example 3: Battery Life

Battery life (hours) from two manufacturers:

Brand A:

11, 12, 13, 12, 11, 14, 13

Brand B:

10, 15, 12, 16, 11, 14, 17

Task: Test whether the variability in battery life differs between the two brands at the 5% significance level.

Example 4: Seed Germination Time

Time (days) taken for seeds to germinate:

Fertilizer A:

7, 8, 9, 8, 7, 8, 9

Fertilizer B:

6, 10, 7, 11, 8, 9, 12

Task: Test whether the two fertilizers produce significantly different variability in germination time.

Example 5: Delivery Time

Delivery times (minutes):

Company A:

32, 31, 33, 34, 32, 31, 35

Company B:

28, 39, 35, 30, 37, 29, 38

Task: At the 5% significance level, test whether the delivery-time variability is the same for the two companies.